

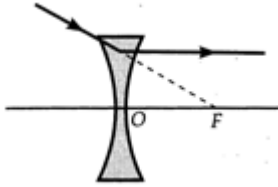
Solution

LIGHT -07

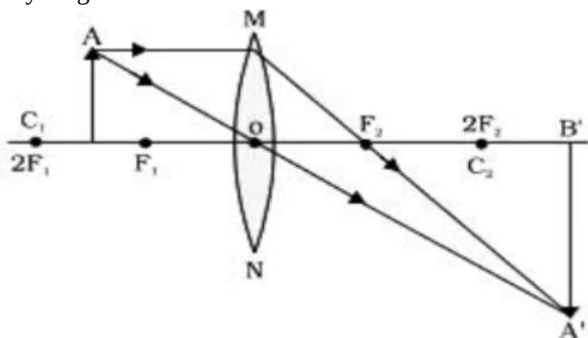
Class 10 - Science

Section A

1. Ray of light after refraction through a concave lens emerges parallel to the principal axis when, the incident ray is directed towards the principal focus of the concave lens.



2. i. The upper part of the mirror is convex because the image appears diminished than its actual size.
ii. The middle part of the mirror is concave as, an enlarged image is formed.
iii. The lower part of the mirror is plane because there is no change in the size of the image.
3. A lens when immersed in liquid becomes invisible. This happens when the refractive index of the liquid is same as that of the lens material.
Under such conditions, no light is reflected by the lens and hence it is not visible.
4. When we read the newspaper the light from sun or an artificial light falls on it. Most of the light except the light that reaches our eyes are absorbed by the newspaper. The light that is not absorbed reaches our eyes which is of the colour's of the newspaper
5. The diver is in the water and the fisherman is on land (i.e., in air). Water is a denser medium than air. It is given that the diver is viewing the fisherman. This indicates that the light rays are traveling from a denser medium to a rarer medium. Hence, the refracted rays will move away from the normal. As a result, the fisherman will appear to be taller.
6. During the summer noon, the air in contact with ground gets heated. When temperature rises, the density of air decreases as the density varies inversely with the temperature. As a result of which there is a movement of hot air towards upward and the cold air downward creating the difference in the densities. So, when light rays pass via it, the path of light rays are changed due to refraction. Due to the fluctuations of air above the ground produce shaking appearance of the tree and houses on the side of an open ground.
7. A concave mirror is preferred as a shaving mirror because it forms a magnified and erect image of the face when the mirror is brought closer to the face. This will make it easy to shave by seeing hair closely and clearly.
8. Yes. The rays incident on a plane mirror when they tend to converge to a point behind the mirror, they have reflected a point on a screen in front of the mirror. Hence, a real image is formed, in this case the object being virtual.
9. Yes, when parallel light is incident on a concave mirror, reflected light converges to its focus. Concave mirror also called as converging mirror.
10. Concave lens can behave as a convex lens when placed in a medium whose refractive index is greater than that of the material of which the concave lens is made.
11. They shine because of total reflection. The speed of light is faster inside the air bubble than in the water. Thus the light gets reflected away from the bubble around the outside, looking like a mirror. This is only along the outer circle in the direction you are looking. If you look towards the center of the bubble you can see right through it.
12. An air bubble in water would be concave lens because the index of refraction of air is less than that of water.
13. Ray diagram:



Here, focal length, $f = 20$ cm, $u = -30$ cm, $h_o = 4$ cm

Using, lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{20} = \frac{1}{v} - \left(\frac{1}{-30} \right)$$

$$\Rightarrow \frac{1}{20} = \frac{1}{v} + \frac{1}{30}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30}$$

Therefore, image distance, $v = 60$ cm

The image formed is real and inverted.

The ratio of the size of the image to the size of the object,

$$\frac{h_1}{h_0} = \frac{v}{u} = \frac{60}{30} = 2$$

14. The image suffers lateral inversion when a plane mirror is placed. The movement of a man, when projected on a screen suffers lateral inversion, whereas the image of a man which is projected on a screen is without lateral inversion.
15. i. As the surface of the screen is rough, light incident on it undergoes reflection making the image visible from all the portions of the audience.
 ii. If it was very smooth, reflected light would reach only a small part of the audience.
 iii. Then the image wouldn't be visible due to its excessive brightness.
 iv. A colored screen would change the color of the image.
 v. A white surface reflects all colors equally producing the image in its actual color.
16. The rays of light travelling parallel to the principal axis, after reflection from a concave mirror meet at a single point only, if the beam of light is narrow or if the mirror is of small aperture, the image formed will be sharper, a smaller aperture mirror will be free from spherical aberration, and produce sharper image.
 In case, a wide beam of light falls on a concave mirror of large aperture, the rays after reflection from the mirror do not come to focus at a single point. Therefore, image formed is not sharper.

Section B

17. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
18. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
19. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Refractive index of diamond w.r.t. liquid

$${}^l\mu_b = \frac{1}{\sin C} = \frac{\mu_d}{\mu_1}$$

$$\frac{\sqrt{6}}{\sqrt{3}} = \frac{1}{\sin C}$$

$$\sin C = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$C = 45^\circ$$
20. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
21. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
22. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
23. (c) A is true but R is false.
Explanation: A is true but R is false.
24. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Linear magnification of a mirror is the ratio of height of the image (cm) and the height of the object (cm) and it has no unit.
25. (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Convex mirrors are not used for rear view on vehicles. We use only concave mirrors for rear view. They will always give erect and diminished image of the object. They provide a wider field for clear view. For this reason, the convex mirrors are fitted on both sides of the vehicles. Therefore, the given assertion is true, but reason is not the correct explanation of assertion.

26.

(c) A is true but R is false.

Explanation: Concave mirrors are used as makeup mirrors because when the face is held within the focus of a concave mirror, then the enlarged image of the face is seen in the concave mirror.

27. (a) Both A and R are true and R is the correct explanation of the assertion.

Explanation: The speeds of light in glass depend on the colour of light because we know that, $v_g = \frac{c}{n_g}$. So, Both A and R are true and R is the correct explanation of the assertion.

28.

(d) A is false but R is true.

Explanation: A is false but R is true.

29. (a) Both A and R are true and R is the correct explanation of A.

Explanation: When the light travels from one medium to another, its speed changes so, that it bends from its path. Therefore, the given assertion is true and given reason is the correct explanation of assertion.

30.

(c) A is true but R is false.

Explanation: A is true but R is false.

31. (a) Both A and R are true and R is the correct explanation of A.

Explanation: A ray of light that travels obliquely from one transparent medium to another will change its direction in the second medium. It is nothing but the process called refraction. This occurs due to change in the speed of light as it enters from one transparent medium to another.

32. (a) Both A and R are true and R is the correct explanation of the assertion.

Explanation: Both A and R are true and R is the correct explanation of the assertion.

33.

(c) A is true but R is false.

Explanation: A is true but R is false. Image is formed behind the mirror as far as object is in front of the mirror. So R is false.

34.

(d) A is false but R is true.

Explanation: A is false but R is true.

35. (a) Both A and R are true and R is the correct explanation of A.

Explanation: When a prism is kept in air, then two angles of incidence will exist for the same magnitude of deviation. This is due to the principle of reversibility of light.

36. (a) Both A and R are true and R is the correct explanation of A.

Explanation: When light ray incident along normal to the mirror, angle of incidence $\angle i = 0^\circ$. According to law of reflection $\angle i = \angle r$, therefore angle of reflection $\angle r = 0^\circ$, i.e. the incident ray retraces its path.

37.

(d) A is true but R is false.

Explanation: A is true but R is false.

38. (a) Both A and R are true and R is the correct explanation of A.

Explanation: According to Snell's law,

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \frac{c/v_2}{c/v_1} = \frac{v_1}{v_2}$$

$$n_1 v_1 = n_2 v_2$$

This shows that the higher is the refractive index of a medium or denser the medium, lesser is the velocity of light in that medium.

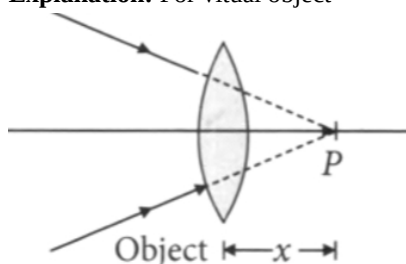
39. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: If object is placed at centre of curvature of concave mirror, its image is at the centre of curvature. Thus, minimum distance between object and its real image is zero. If object is between pole and focus of concave mirror, its image is virtual and magnified.
40. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The Refractive index of any pair of media is inversely proportional to the wavelength of light.
Hence, $\gamma_v < \gamma_r$
 $\mu_r < \mu_v$
where, γ_v and γ_r are the wavelengths of violet and red light. μ_v and μ_r are the refractive index of violet and red light.
41. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
42. **(d)** A is false but R is true.
Explanation: We can see the virtual image when the object is in between f and P.
43. **(a)** Both A and R are true and R is the correct explanation of the assertion.
Explanation: According to new sign conventions, the height of an object above the principal axis is always taken as positive.
44. **(a)** Both A and R are true and R is the correct explanation of the assertion.
Explanation: plane mirror has a property to form images whose distance from the mirror is the same as that of the object from the mirror
so we can say that
distance of object = u
distance of image = -u (since the image formed will be on the opposite side)
So, the distance between the image and the object will be 2u
Let the object travels with speed 2m/s towards the mirror
so in unit time distance travelled will be 2m
the new separation will be 2(u-2)
We can say that the distance reduced in unit time between the object and the mirror is 2×2
hence the speed of the image will be 2×2 = 4m/s
45. **(c)** A is true but R is false.
Explanation: A is true but R is false.
46. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: The plane mirror forms a virtual image of the real object and the real image of the virtual object.
47. **(a)** Both A and R are true and R is the correct explanation of the assertion.
Explanation: A convex mirror is used as a driver mirror for rearview because it has a much larger field view as compared to plane mirror.
48. **(c)** A is true but R is false.
Explanation: A convex lens made of glass behaves as a convergent lens when placed in air or water. However when the same lens is immersed in carbon disulphide ($n = 1.63$), it behaves as a divergent lens. Therefore when a convergent lens is placed inside a transparent medium of refractive index greater than that of material of the lens, it behaves as a divergent lens. Behaviour of a lens depends on the refractive index of a surrounding medium.
49. **(d)** A is false but R is true.
Explanation: A is false but R is true.
50. **(d)** A is false but R is true.
Explanation: If the rays cross focal point of convex lens, they become diverging.

51. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: We know that the refractive index of Kerosene is 1.44 and the refractive index of water is 1.33. It shows that kerosene is optically denser than water. But the mass density of kerosene is lesser than water. So, there is no relation between mass density and optical density. Therefore, it clearly defines that an optically denser medium may not possess greater mass density. Thus, the given assertion and reason are true but reason is not the correct explanation of assertion.
52. **(c)** A is true but R is false.
Explanation: An image on a plane mirror is virtual and it can be photographed.
53. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
54. **(d)** A is false but R is true.
Explanation: The size of the image does not affect the nature of the image except that a bigger image as it gathers more light rays due to wider aperture.
55. **(c)** A is true but R is false.
Explanation: The object is placed between F and 2F of the objective lens.
56. **(c)** A is true but R is false.
Explanation: Diffuse reflection is caused by the roughness (or irregularities) in the reflecting surface of an object. The laws of reflection are valid at each point even on the rough surface of an object.
57. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Optical fiber communication is based on the phenomenon of total internal reflection at the core-clade interface. The refractive index of the material of the cladding, hence, light sinking at core-cladding interface gets totally internal reflected. The light undergoes and reaches the other end of the fiber.
58. **(c)** A is true but R is false.
Explanation: A is true but R is false.
59. **(c)** A is true but R is false.
Explanation: Angle of incidence = Angle between incident ray normal to the mirror = 0°
60. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
61. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Concave mirror converges the light rays falling on it to a point. So large concave mirrors are used to concentrate sunlight to produce heat in solar cookers.
62. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
63. **(c)** A is true but R is false.
Explanation: The image of a real object may be real in the case of a concave mirror.
64. **(c)** A is true but R is false.
Explanation: A is true but R is false.

65.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation: For virtual object



$$u = +x, F = +f$$

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{F}$$

$$\text{or } \frac{1}{v} - \frac{1}{x} = \frac{1}{f} \text{ or } \frac{1}{v} - \frac{1}{f} = \frac{1}{x}$$

$\therefore v > 0$, it means image is real.

66.

(c) A is true but R is false.

Explanation: Since the lens is made of two different materials of different refractive indices, there will be two different focal lengths of the lens. Hence two images will be formed. The image formed by convex lens is always real except in case when object is placed between optical centre and focus.

67.

(d) A is false but R is true.

Explanation: The focal length of the spherical mirror does not depend on the medium in which it is placed.

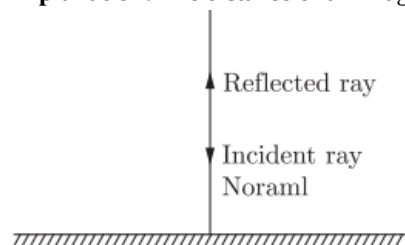
68. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: When the object is placed at an infinity position, the image formed by a concave mirror will be placed at the focus. Then the size of the image will be highly diminished and point-size. So, the nature of the image formed by the concave mirror is real and inverted.

69.

(d) A is false but R is true.

Explanation: The distance of an image in a convex mirror is always finite.



Angle of reflection = 0° (from laws of reflection)

Hence, the reflected ray retraces its path along the normal at an angle 0° with normal.

70.

(c) A is true but R is false.

Explanation: In concave spherical mirrors, the real image may also form when object is placed in between F and P.

71. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.